



DPR: MSK: Wrist and Hand

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Principles and Practices of Osteopathic Medicine 1



SESSION OBJECTIVES

THE STUDENT PHYSICIANS WILL BE ABLE TO:

1. **Demonstrate** the components of the **history and physical** examination of the wrist and hand
2. **Evaluate** the patient utilizing **inspection, palpation, active range of motion, passive range of motion, strength (motor), neurologic, vascular, and osteopathic examination** of the wrist and hand
3. **Apply** testing and signs of the wrist and hand including **Phalen's test, Tinel's sign, thumb abduction, Allen test, and the Finklestein test**
4. **Distinguish normal versus abnormal examination** findings.
5. **Anticipate** and discuss common abnormal examination findings including **carpal tunnel syndrome, scaphoid fracture, and De Quervain's Tenosynovitis, and Arthritis.**
6. Recall the appropriate way to **document** common normal and abnormal findings of history and examination intake by **writing a SOAP note within 9 minutes.**

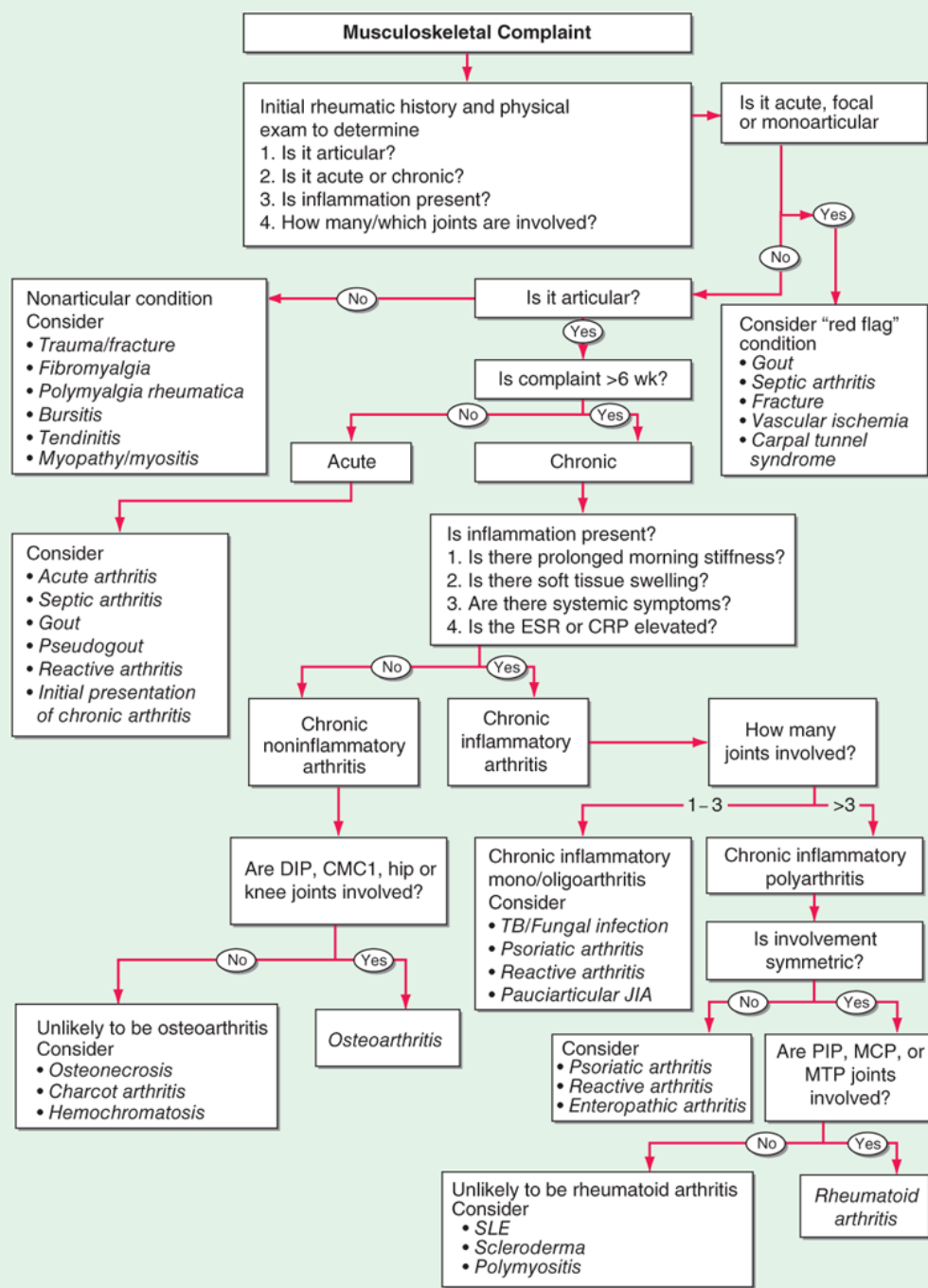
What are we talking about today?

- History considerations for the MSK complaint (and wrist/ hand)
- Physical exam of the wrist and hand
- Testing for abnormalities of wrist and hand
- Osteopathic Treatments
- SOAP note writing

- **The mystery is in the history**
 - What could it be?
- **Get quizzical with your physical**
 - Why am I doing this? Is there a more specific exam?

MSK Overview

ALGORITHM FOR MUSCULOSKELETAL COMPLAINTS



Algorithm for diagnosis of musculoskeletal (eds). Chapter 331, Approach to Articular New York, NY: McGraw-Hill; 2012.

- Must ask and fully understand the...

MECHANISM OF INJURY

Wrist and Hand – HISTORY!

- Location(s)?
- Unilateral or bilateral?
- Pain – grade it from 0-10. Does it radiate?
- Swelling – currently present or recently present???

Wrist and Hand – HISTORY!

- Onset – abrupt or insidious?
- Progression- what has been going on since the onset?
- Paresthesia – any numbness or tingling?
Where? Which fingers?

Wrist and Hand – HISTORY!

- Any aggravating activities??
- Weakness – difficulty writing, cannot turn a door knob, cannot do a push up, cannot type, etc.
- Any repetitive hobbies like sewing, painting, archery, or underwater basket weaving???

LIFE LESSON TIME!!!

- After asking your history questions...
 - Time to think about what your potential diagnosis list will look like
- How will this list look after you perform your physical exam?
 - Will things be ruled in or excluded?
 - What other data will you need from testing?

Assessing the Four Signs of Inflammation

- **Swelling (Tumor)**. Palpable swelling may involve: (1) the synovial membrane, which can feel boggy or doughy; (2) effusion from excess synovial fluid within the joint space; or (3) soft tissue structures, such as bursae, tendons, and tendon sheaths.
 - Palpable **bogginess or doughiness** of the synovial membrane indicates *synovitis*, which is often accompanied by effusion. Palpable joint fluid is present in effusion, tenderness over the tendon sheaths in *tendinitis*.
- **Warmth (Calor)**. Use the backs of your fingers to compare the involved joint with its unaffected contralateral joint, or with nearby tissues if both joints are involved.
 - Increased warmth is seen in arthritis, tendinitis, bursitis, and *osteomyelitis*.
- **Redness (Rubor)**. Redness of the overlying skin is the least common sign of inflammation near the joints and is usually seen in more superficial joints like fingers, toes, and knees.
 - Redness over a tender joint suggests septic or crystalline arthritis, or possibly RA.
- **Pain or tenderness (Dolor)**. Try to identify the specific anatomic structure that is tender.
 - Diffuse tenderness and warmth over a thickened synovium suggest arthritis or infection; focal tenderness suggests injury and trauma.

Wrist and Hand – EXAM!

- Inspection
- Palpation
- Active Range of Motion (AROM)
- Passive Range of Motion (PROM)
- Strength Testing
- Abnormality Confirmation Testing
- Neurovascular Exam
- Osteopathic Exam/ Treatment

Wrist and Hand – INSPECTION!

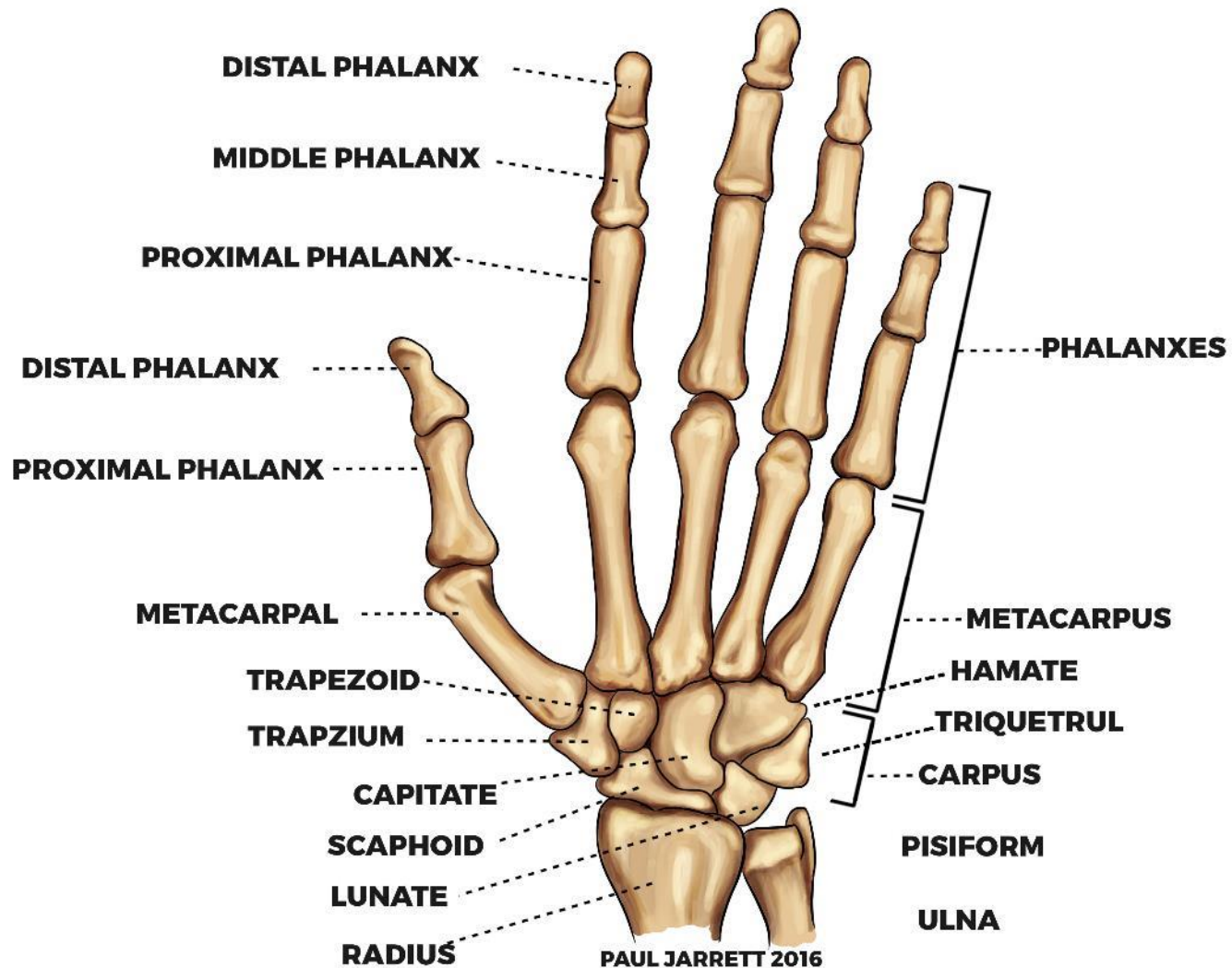


Wrist and Hand – INSPECTION!



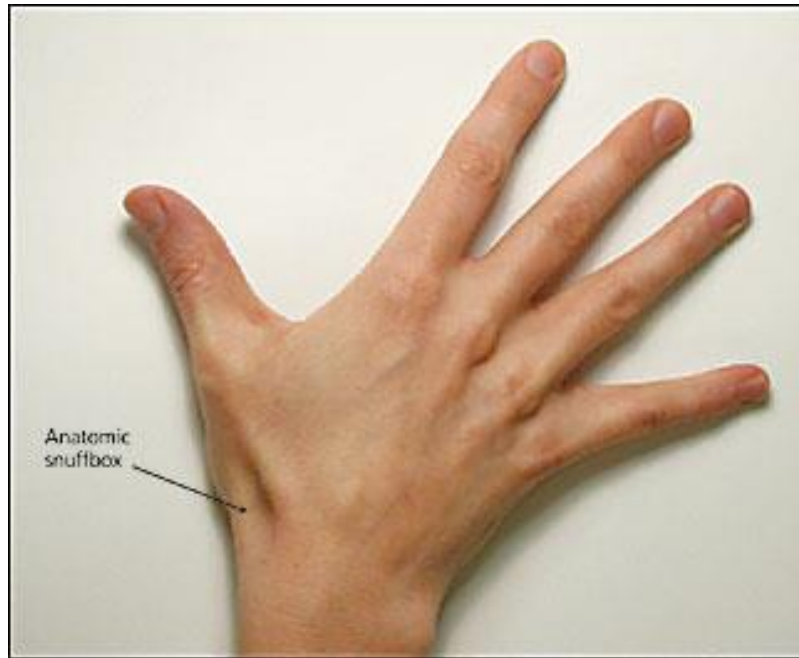
Wrist and Hand - Anatomy

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- Palpate key landmarks
- Concentrate on any abnormal areas noted in the inspection
- BE SYSTEMATIC
 - Don't jump around- you will miss important findings

KEY landmark – Scaphoid bone



Wrist and Hand - PALPATION

- KEY landmark – base of the thumb



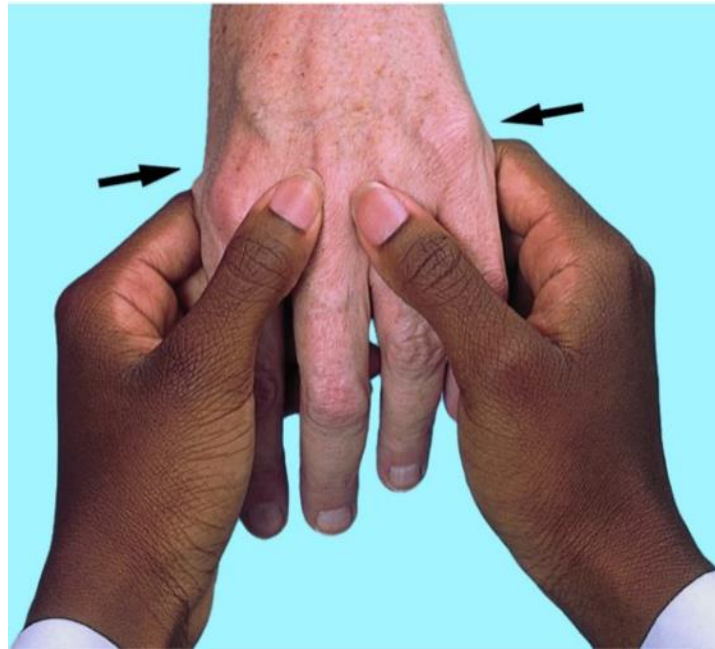
Wrist and Hand - PALPATION

- KEY Landmark – hook of the hamate



Wrist and Hand - PALPATION

- KEY landmark – MCP joints (possible area of inflammation)



Wrist and Hand - PALPATION

- Palpating the Wrist



Bates' Guide to Physical Examination and History Taking, 12e, 2017

Wrist and Hand– ACTIVE Range of Motion

- Make a fist



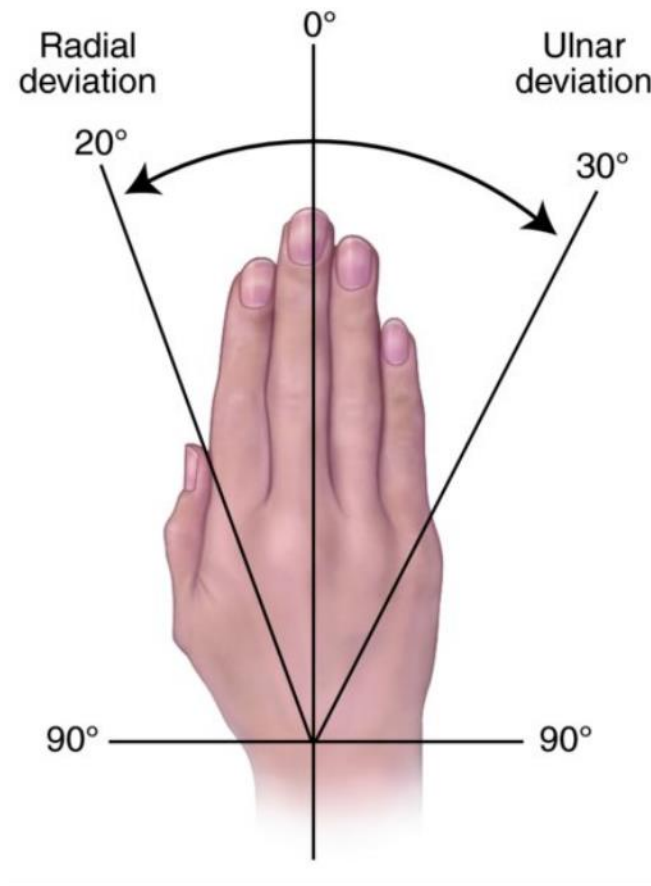
- Spread the fingers



Compare each side!!!

Wrist and Hand– ACTIVE Range of Motion

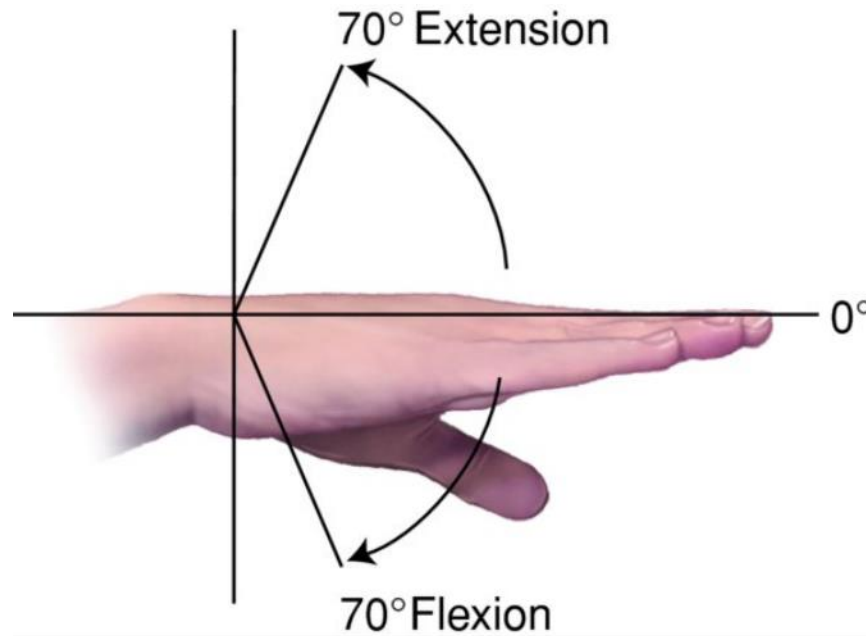
- Wrist Radial and Ulnar Deviation



Bates' Guide to Physical Examination and History Taking, 12e, 2017

Wrist and Hand– ACTIVE Range of Motion

- Wrist Flexion and Extension



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- **BREAK TIME!**
 - Stand up Stretch Move around



- Assess whether ROM is limited by pain or swelling
- Do this for any abnormal joint found during active range of motion

Wrist and Hand – Strength Testing

- Graded on scale out of 5...
- 5/5 – full strength
- 4/5 – diminished strength
- 3/5 – movement against gravity
- 2/5 - movement without gravity
- 1/5 - involuntary movements
- 0/5 - no movement at all!!!

Wrist and Hand – Grip Strength

- Tests intrinsic hand muscles and finger flexors...

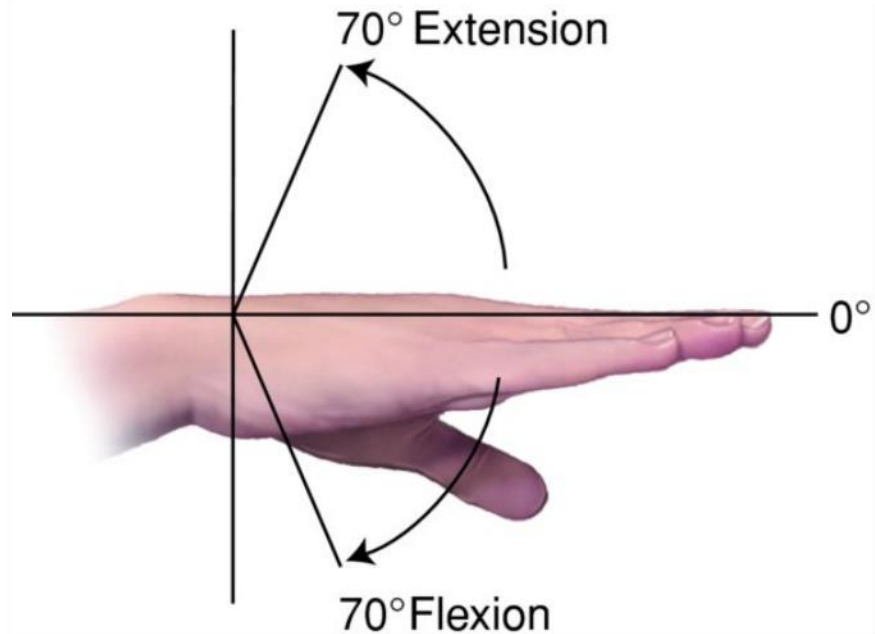
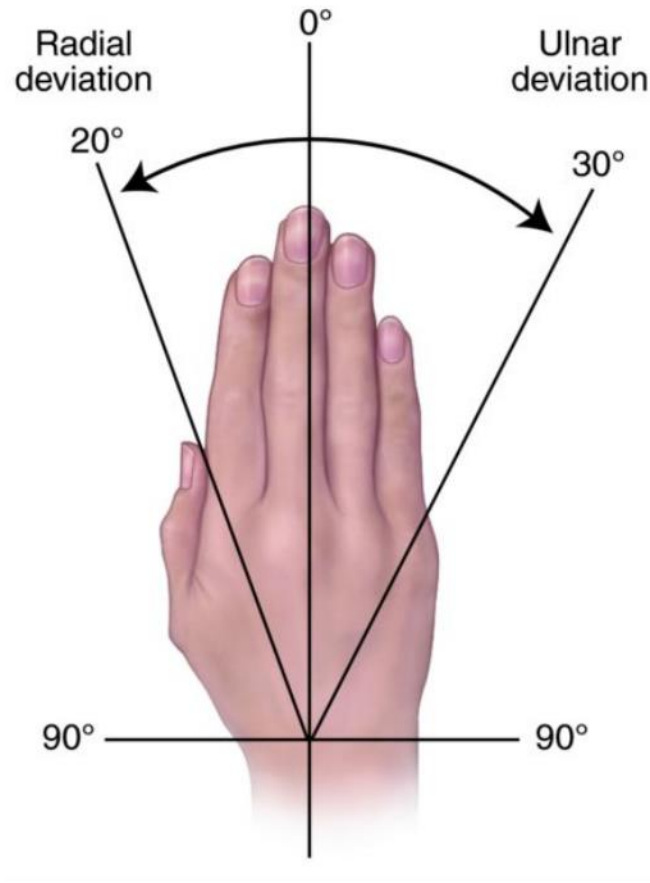


- Decreased grip strength is a **positive test** for weakness of the finger flexors and/or intrinsic muscles of the hand.
- Grip weakness plus wrist pain are often present in *de Quervain tenosynovitis*.

- Test individual digits if there is an injury to that digit
- Use minimal resistance
- **Compare to the other side!!!**

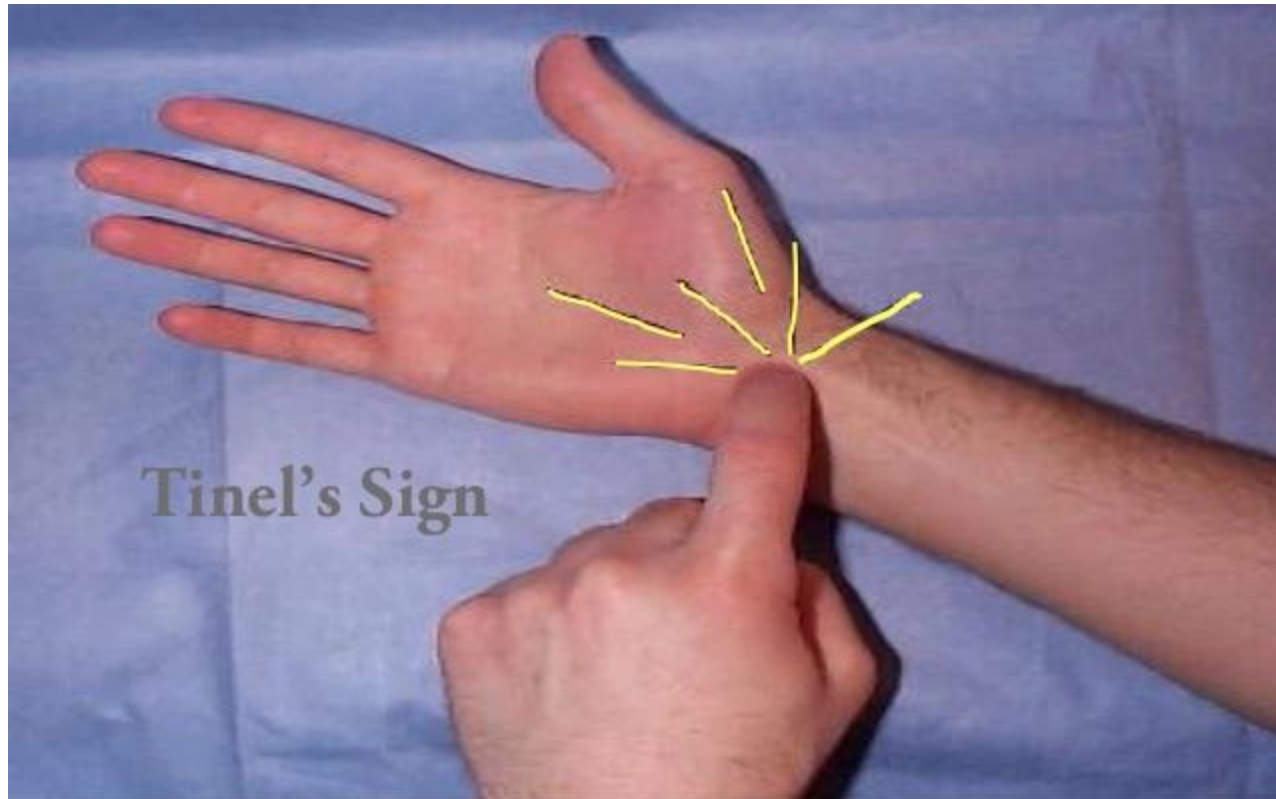
Wrist and Hand – Wrist Strength

- “Resisted Range of motion”



- TINEL's – tapping over the median nerve
 - Reproduces pain and paresthesias in the distribution of the **median nerve**
- Tests for Carpel Tunnel Syndrome (CTS)

Tinel's Test



- **Phalen's Test** – compressing the median nerve
 - Hyper-flex both wrists and hold the dorsal hands together for 30-60 seconds
- Monitor for pain or paresthesias in the distribution of the **median nerve**

Phalen's Test



- **Finklestein's Test** – stressing the Extensor Pollicus Brevis (EPB) and Abductor Pollicus Longus (APL)
 - Fold the thumb into the hand and then adduct the hand with some force
- Test for **DeQuervain's Tenosynovitis**

Finklestein's Test



- **Thumb Abduction Test** - extend the thumb. Have patient abduct the thumb against resistance.
 - If there is weakness present, it may indicate Carpal Tunnel Syndrome.
- Why???

In the event of a serious injury, you must assess integrity of the nerves and blood supply!!

- Check Pulses
 - Radial Artery
 - Ulnar Artery may be too deep to palpate

Pulses are grade on 4 point scale...
2/4 is normal.

- Check capillary refill in several fingers
 - Compress the nail bed of a digit for 2-3 seconds. Release pressure, and immediately observe the return of circulation (pink color)
- Normal time is less than 3 seconds
- Maybe compare to other side if unsure

- Allen Test – checking for adequate perfusion to the hand
 - Occlude both the radial and ulnar arteries. Have patient make a fist/open hand several times. Release occlusion over radial artery. Monitor hand color.
- **Repeat** for Ulnar Artery.

- Do this before starting an “arterial line” in the radial artery...



- Check light touch sensation in multiple regions of hand and wrist
 - C6, C7, C8 dermatomes
- If light touch is questionable, then test sharp sensation
 - Break in half a tongue depressor for sharp tool

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- SAMPLE EXAM VIDEO (watch before lab)
 - [Wrist and Hand Exam](#)

- 46 year old female presents with **right wrist** pain. Pain **started 10 days ago, gradually**, and has **progressively worsened**.

Pain is a **tingling numbness** sensation in the **middle of her palm**. Pain rated **4/10** currently, 7/10 at worst, 3/10 at best. She has tried hand **stretches, ice, heat, 200 mg ibuprofen** daily without significant improvement.

Typing at work **makes her pain worse**. She **denies radiation** of pain.

- **What is on your differential list?**

- **Physical exam:**
- Gen: Alert, Oriented to person, place, time; No apparent distress. Well groomed.
- MSK:
 - Inspection: No gross abnormalities appreciated
 - Tenderness: Located to midline wrist right hand
 - ROM: Non-affected digits show full passive and active range of motion
 - Tendons: Non-affected digits glide freely without evidence of lag or incompetence or triggering
 - Neurovascular: Symmetric pulses and sensation bilaterally; Decreased grip strength right hand
 - Provocation tests: Positive Tinel's, Positive Phalen's
- Osteopathic: Decreased motion to lunate, capitate of right hand

WHAT NOW?



Osteopathic Manipulative Medicine for Carpal Tunnel Syndrome

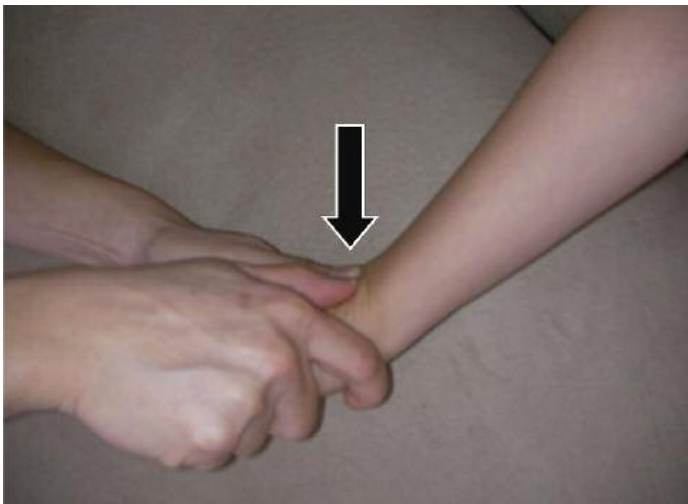
Gilbert Siu, DO, PhD; J. Douglas Jaffe, DO; Maryum Rafique, DO, MA; and Michael M. Weinik, DO, MS

Wrist and Hand – Osteopathic Exam and Treatment



<http://jaoa.org/article.aspx?articleid=2094358>

Wrist and Hand – Osteopathic Exam and Treatment



Wrist and Hand – Documentation (Comprehensive)

Objective MSK

Observation: Pt. was point tender over carpal tunnel on R wrist w/o crepitation. There was a little bit of swelling in the wrist but not a whole lot. No obvious deformity was found, however, the R thenar eminence was smaller due to atrophy of the muscles.

<u>PROM/AROM:</u> AROM:	Wrist Flexion: 83	Finger flexion(MCP): 70
	Wrist Extension: 87	Finger extension(MCP):43
	Wrist radial deviation: 20	Finger abduction: 25
	Wrist ulnar deviation: 30	Finger adduction: 23
PROM:	Wrist Flexion: 85	Finger flexion(MCP): 75
	Wrist extension: 90	Finger extension(MCP): 47
	Wrist radial deviation: 27	Finger abduction: 28
	Wrist ulnar deviation: 33	Finger adduction: 25
<u>RROM/MMT:</u>	Wrist flexion: 5/5	Finger adduction: 5/5
	Wrist extension: 5/5	Thumb MP and IP flexion: 5/5
	Finger MP flexion: 5/5	Thumb MP and IP extension: 5/5
	Finger PIP and DIP flexion: 5/5	Thumb abduction: 5/5
	Finger MP extension: 5/5	Thumb adduction: 5/5
	Finger abduction: 5/5	

Neurological Testing (Reflexes, dermatomes, myotomes):

Dermatomes- C6, C7, C8 (All WNL)
Reflexes- Biceps (C5-C6) (WNL)
Brachioradialis (C6) (WNL)
Triceps (C7) (WNL)

Cardiovascular Testing:

Radial artery: 2/4
Ulnar artery: 2/4

Special Tests: Tinel's sign at wrist (+), Phalens (+),

Osteopathic: As above, Decreased motion to lunate, capitate of right hand

Hand and Wrist - References

- Bates, Guide to Physical Examination and History Taking, Chapter 16, The Wrist and Hand.
- Netter's Concise Orthopaedic Anatomy, 2nd Edition, Chapter 6, Hand.

(both are located on the NYITCOM Library website)

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- Questions?
 - Contact info:
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